

WEEK 7 PROJECT SUMMARY

1. What was planned to be tested

The planned work was to test the Week 6 candidate model, evaluate its overall predictive performance, analyse false positives and false negatives, examine performance across important segments, assess robustness and generalization, test targeted refinements, conduct re-testing, and complete at least one meaningful cross-track validation activity.

2. What was actually tested

The Week 6 Refined Logistic Regression model was re-established and tested using the same held-out test structure.

Testing covered:

- Overall classification metrics,
- Confusion matrix,
- False-positive and false-negative patterns,
- Booking lead-time groups,
- Previous no-show history,
- Appointment type,
- Age groups,
- Appointment day,
- Appointment time,
- Distance groups,
- Training-versus-test performance,
- Existing 5-fold cross-validation results,
- A logarithmic booking lead-time feature,
- A Data Analytics-informed additive lead/no-show score.

3. Most important testing results

The retained Week 6 candidate achieved:

Metric	Result
Accuracy	61.50%
Precision	60.87%
Recall	69.28%
F1-score	64.80%
ROC-AUC	66.47%

The confusion matrix was:

[[247, 216],

[149, 336]]

This corresponds to:

- True Negatives: 247
- False Positives: 216
- False Negatives: 149
- True Positives: 336

The test-set results were reasonably consistent with Week 6 5-fold cross-validation results, although overall predictive performance remained moderate.

4. Issues or weaknesses Identified

The main weaknesses identified were:

- Moderate overall predictive performance,
- Continued false-positive and false-negative errors,
- Variation in performance across booking lead-time groups,
- Relatively low recall among some shorter booking lead-time groups.

No strong independent weakness was identified that justified changing the model solely on the basis of appointment type, age, appointment day, appointment time or distance.

5. Improvements and refinements made

A targeted refinement was developed by adding a logarithmic representation of booking lead time:

$\text{booking_lead_log} = \log_{10}(\text{booking_lead_days})$

This refinement was motivated by the ordered relationship between booking lead time and no-show behaviour identified through the Data Analytics findings. The Data Analytics recommendation concerning the combined effect of booking lead time and previous no-show history was also operationalized as an additive `lead_no_show_score` for collaborative testing.

6. Retesting results

The `booking_lead_log` refinement produced almost identical overall performance:

Metric	Week 6 candidate	Week 7 refinement
Accuracy	61.50%	61.50%
Precision	60.87%	60.91%
Recall	69.28%	69.07%
F1-score	64.80%	64.73%
ROC-AUC	66.47%	66.52%

The refinement did not consistently improve performance across the booking lead-time groups and was therefore rejected. The retained Week 6 candidate was subsequently re-tested and reproduced its original metrics exactly. This confirmed reproducibility on the same held-out test set.

7. Tracks collaborated with

The main cross-track collaboration was with the Data Analytics track. The Data Analytics team provided validated findings concerning booking lead time and previous no-show history. In return, the Data Science track provided the model feature set, candidate model evaluation, confusion matrix, segment-level findings and refinement outcomes.

8. What was tested collaboratively

The collaborative test examined whether the Data Analytics finding that booking lead time and previous no-show effects were additive could be represented through a simple summed lead_no_show_score. The score combined ordinal booking lead-time groups with previous no-show groups. The resulting Logistic Regression model was compared with the retained Week 6 candidate using the same evaluation framework.

9. What changed as a result

The additive score did not improve the candidate model.

Metric	Week 6 candidate	Collaborative model
Accuracy	61.50%	61.18%
Precision	60.87%	60.73%
Recall	69.28%	68.25%
F1-score	64.80%	64.27%
ROC-AUC	66.47%	66.39%

The additive representation was therefore not adopted. The Week 6 Refined Logistic Regression remained the retained candidate model. The unsuccessful refinement was documented as evidence rather than discarded.

10. Key findings and validation outcomes

The testing confirmed that the retained Logistic Regression model is reasonably stable across the current evaluation framework but has moderate predictive performance. Booking lead time remained the clearest segment-level area requiring attention, while previous no-show history remained an important modelling feature. The cross-track testing also demonstrated that a valid analytical finding does not necessarily mean that every proposed feature representation will improve predictive performance. The Week 7 process therefore provided evidence supporting the decision to retain the existing candidate model.

11. Major challenges encountered

The main challenges were:

- Interpreting variation across multiple segments without overreacting to small subgroup sizes,
- Distinguishing meaningful refinement from negligible metric changes,
- Ensuring that proposed features did not introduce prediction-time leakage,
- Translating Data Analytics findings into a testable modelling representation,
- Maintaining continuity with the established Week 6 modelling workflow.

12. Important decisions made

The following decisions were made during Week 7:

1. Retain Week 6 Refined Logistic Regression as the candidate model,
2. Reject the booking_lead_log refinement because it did not provide consistent improvement,
3. Reject the Data Analytics-informed additive lead_no_show_score because its performance was slightly lower than the retained candidate,
4. Retain the unsuccessful tests as documented evidence supporting the modelling decision,
5. Continue monitoring booking lead-time segments during subsequent validation.

13. Remaining limitations

- The model has moderate predictive performance and continues to produce false positives and false negatives,
- The current evaluation uses the available fictional HealthConnect dataset and does not establish performance on an independent or temporally separated real-world dataset,
- Probability calibration and alternative classification thresholds have not yet been fully evaluated,
- The relationships identified in the data are predictive associations and should not be interpreted as evidence of causal relationships.

14. Remaining issues or dependencies

Further validation remains necessary before any production use is considered.

Remaining dependencies include:

- Final model integration requirements from the ML Engineering track,
- Additional validation using independent or temporally separated data where available,
- Classification threshold testing,
- Probability calibration,
- Continued monitoring of booking lead-time segments,
- Final cross-track integration requirements,
- Documentation and repository updates.

Any future operational use would also require appropriate human oversight and monitoring.

15. Contribution to the overall HealthConnect project

The Data Science contribution provided a tested and reproducible candidate model for appointment no-show prediction, supported by:

- Error analysis,
- Segment-level evaluation,
- Robustness testing,
- Targeted feature refinement,
- Re-testing,
- Cross-track validation,
- Evidence-based model selection.

The work also provided documented evidence showing which proposed feature changes should not be adopted. This strengthens the wider HealthConnect solution by providing a documented modelling component that can proceed to final integration and validation.

16. What must be completed before Week 8

Before Week 8, the project should complete the remaining model integration and validation requirements.

Priority activities include:

1. Confirm the final candidate feature set and model interface,
2. Conduct additional independent or temporal validation where possible,
3. Evaluate classification thresholds,
4. Evaluate probability calibration,
5. Confirm cross-track requirements with ML Engineering,
6. Document the final testing evidence,
7. Update the repository and README where applicable,
8. Finalize the model documentation,
9. Prepare the supporting materials required for Week 8 final integration and presentation.

17. Overall Week 7 outcome

Week 7 successfully moved the HealthConnect Data Science contribution through the required progression:

Test → Identify Issues → Refine → Re-test → Validate → Document → Prepare for Week 8

The principal outcome was not a numerical performance increase, but an evidence-based validation of the retained candidate model and a documented decision not to adopt refinements that did not demonstrate meaningful improvement. The Week 6 Refined Logistic Regression therefore remains the current candidate for the next stage of HealthConnect development, subject to further validation and final integration.